№3.

Рассм. $\widetilde{\theta}_1$:

a) $M[\overline{X}] = M[\frac{1}{n}\sum_{i=1}^{n} x_i] = \frac{1}{n}\sum_{i=1}^{n} M[x_i] = M[\xi] = \theta$

т.о. $\widetilde{\theta}_1$ – несмещ.

b) $D[\widetilde{\theta}_1] = D[\frac{1}{n}\sum_{i=1}^{n} x_i] = \frac{1}{n^2}\cdot nD\xi = \frac{\theta^2}{n}$

при $n=3 \longrightarrow D[\widetilde{\theta}_1] = \frac{\theta^2}{3}$

Рассм. $\widetilde{\theta}_2$:

a) $\varphi(y) = n\frac{1}{\theta}\cdot C_{n-1}^{1}\left(1-\left(1-\exp\left(-\frac{x}{\theta}\right)\right)^{n-2}\left(1-\exp\left(-\frac{x}{\theta}\right)\right)\right) =$

$= \frac{n(n-1)}{\theta}\left(\exp\left(-\frac{y}{\theta}(n-1)\right) - \exp\left(-\frac{y}{\theta}n\right)\right)$

т.о. $M[\widetilde{\theta}_2] = \int_{-\infty}^{\infty} y\,\varphi(y)\,dy = \int_{0}^{+\infty} \frac{n(n-1)}{\theta} y\left(\exp\left(-\frac{y}{\theta}(n-1)\right) - \right.$

$\left. - \exp\left(-\frac{y}{\theta}n\right)\right)dy = \frac{n(n-1)}{\theta}\int_{0}^{+\infty}\frac{y}{\exp\left(\frac{y}{\theta}n\right)}\left(\exp\left(\frac{y}{\theta}\right)-1\right) = \frac{2n-1}{n^2-n}\theta$

Введём $\widetilde{\theta}_2'$ – исправл.

$\widetilde{\theta}_2' = \frac{n(n-1)}{2n-1}\cdot\widetilde{\theta}_2 = \frac{n(n-1)}{2n-1}\cdot X_{(2)}$

b) $M[\widetilde{\theta}_2] = \int_{0}^{+\infty}\frac{n(n-1)}{\theta}y^2\left(e^{-\frac{y}{\theta}(n-1)} - e^{-\frac{y}{\theta}n}\right)dy = 2\theta^2\left(\frac{3n^2-3n+1}{n^2(n-1)^2}\right)$

т.о. $D[\widetilde{\theta}_2] = \theta^2\left(\frac{2n^2-2n+1}{n^2(n-1)^2}\right)$ и

$D[\widetilde{\theta}_2'] = \frac{n^2(n-1)^2}{(2n-1)^2}D[\widetilde{\theta}_2] = \theta^2\left(\frac{2n^2-2n+1}{4n^2-4n+1}\right)$

при $n=3 \longrightarrow D[\widetilde{\theta}_2'] = \frac{13}{25}\theta^2$

b) Итог:

$D[\hat{\theta}_2'] > D[\hat{\theta}_1] \Rightarrow \hat{\theta}_1$ - более эфф.

c) Докажем рег. П.В.М.:

$$p(x) = \begin{cases} \frac{\exp(-\frac{x}{\theta})}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$, где пар. $\theta > 0$

1) $p(x)$ - непр. дифф. по θ на $(0; +\infty)$

2) $\frac{\partial}{\partial\theta}\left(\int_0^{+\infty} \frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right)dx\right) = \frac{\partial}{\partial\theta} 1 = 0$

$\int_0^{+\infty}\left(\frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right) - \frac{1}{\theta^2}\exp\left(-\frac{x}{\theta}\right)\right)dx = \frac{1}{\theta} + 0 - \frac{1}{\theta} = 0$

$\Rightarrow$ рав-во соблюд.

3) $\ln\left(\frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right)\right) = -\frac{x}{\theta} - \ln\theta$

$\left(\frac{\partial \ln p}{\partial\theta}\right)^2 = \left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2$

$\Rightarrow I(\theta) = M\left[\left(\frac{\partial \ln p}{\partial\theta}\right)^2\right] = \int_0^{+\infty}\left(\left(\frac{x}{\theta}\right)^2 - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right)\frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right)dx =$

$= \frac{1}{\theta^2}$ - $\left.\begin{matrix}\text{непр.} \\ >0\end{matrix}\right\}$ на $(0; +\infty)$

$\Rightarrow$ П.В.М. регул.

Рассм. рег. оценок

$$\hat{\theta}_1 : \begin{cases} \text{несмещ. оценка} \\ \text{ПВМ-рег.} \\ D[\hat{\theta}_1] = \frac{\theta^2}{n} \text{ - опр. на } \forall \text{ комп. на } (0; +\infty) \text{ по } \theta \end{cases}$$

$\Rightarrow \widetilde{\theta}_1$ – рег. оценка

$\widetilde{\theta}_2^{\,1}$: $\begin{cases} \text{несмещ. оценка} \\ \text{ПВМ – регул.} \\ D[\widetilde{\theta}_2^{\,1}] = \theta^2\left(\frac{2n^2-2n+1}{4n^2-4n+1}\right) \text{ опр. на } \forall \text{ компакте на } (0;+\infty) \text{ по } \theta \end{cases}$

$\Rightarrow \widetilde{\theta}_2^{\,1}$ – регул. оценка

$\widetilde{\theta}_1$: $\begin{cases} \text{ПВМ-регул} \\ \widetilde{\theta}_1\text{-регул} \\ \theta\text{-дифф} \end{cases} \Rightarrow \forall\, \theta \in (0;+\infty) \hookrightarrow$

$\hookrightarrow D[\widetilde{\theta}_1] \geq \frac{\theta^2}{3}$

$D[\widetilde{\theta}_1] = \frac{\theta^2}{3} \Rightarrow \widetilde{\theta}_1$ – эфф.

$\widetilde{\theta}_2^{\,1}$: $\begin{cases} \text{ПВМ-регул.} \\ \widetilde{\theta}_2^{\,1}\text{-регул} \\ \theta\text{-дифф.} \end{cases} \Rightarrow \forall\, \theta \in (0;+\infty) \hookrightarrow$

$\hookrightarrow D[\widetilde{\theta}_2^{\,1}] \geq \frac{\theta^2}{3}$

$D[\widetilde{\theta}_2^{\,1}] \neq \frac{\theta^2}{3} \Rightarrow \widetilde{\theta}_3^{\,1}$ – не эфф. и т.к. по т. ед. эфф. оценки

$\begin{cases} \exists\,!\, \widetilde{\theta}_1 \text{ эфф.} \\ \widetilde{\theta}_2 \neq \widetilde{\theta}_1 \end{cases} \Rightarrow \widetilde{\theta}_2$ не эфф.